

Project Report

Import Data using Transform Maps (Spreadsheet) – ServiceNow

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1. INTRODUCTION

1.1 Project Overview

This micro project imports structured employee data from an external spreadsheet into the ServiceNow platform using Import Sets and Transform Maps. Bulk employee data received in Excel format is loaded into a staging table (u_employee_import), mapped to a custom target table (Employee Test, u_employee_test) through a Transform Map, and validated. Coalesce on Employee ID prevents duplicate records on repeated imports, and three reports on an Employee Analytics Dashboard give a single view of the employee data.

1.2 Purpose

- Migrate bulk employee data into ServiceNow efficiently and accurately.
- Understand the end-to-end data import process: data loading, field mapping, transformation execution and result verification.
- Avoid duplicate records when the same data is imported more than once.
- Provide visual insight into employee data through reports and a dashboard.

2. IDEATION PHASE

2.1 Problem Statement

Organizations receive bulk employee data in Excel format. Entering it into ServiceNow manually is slow and error-prone, and re-importing the same file creates duplicate employee records. HR and admin teams also lack a single view of employee data and import quality. A reliable, automated import with duplicate control and reporting is needed.

2.2 Empathy Map Canvas

Says	Thinks
"Entering this list record by record takes too long." "Please check there are no duplicates."	Will the data be accurate? Will a second upload overwrite or duplicate records?
Does	Feels
Maintains employee lists in Excel/Google Sheets, shares files by email, enters data manually and checks lists afterwards.	Frustrated by repetitive work and worried about errors; confident and in control when records are validated.

2.3 Brainstorming

Ideas considered for handling the employee data:

- Manual entry of each record in ServiceNow (slow, error-prone).

- Custom import script (needs coding and maintenance).
- Import Set with Transform Map (low-code, reusable, validated through Transform History) – selected.
- Coalesce on Employee ID to update existing records and avoid duplicates – selected.
- Reports and a dashboard for visibility – selected.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Stage	Customer action	Experience	Opportunity
Receive data	HR receives the employee Excel sheet	Unstructured, may contain changes	Standardize the sheet columns
Load	Admin uses Load Data to stage the file	Quick, errors visible in the import log	Import Set staging table
Transform	Admin maps fields and runs the Transform Map	Records created in the target table	Mapping Assist, Coalesce
Validate	Admin checks Transform History and Employee Tests	Confident that inserts/updates are correct	Counts of inserts, updates, ignored
Review	HR opens the dashboard	Clear picture of departments and locations	Reports and dashboard

3.2 Solution Requirement

Functional requirements:

- Load an Excel file (sheet 1, header row 1) into an Import Set table.
- Create the custom table Employee Test with Employee ID, Employee Name, Email, Department and Location.
- Map source fields to target fields with a Transform Map and run the transform.
- Coalesce on Employee ID so that existing records are updated and only new records are inserted.
- Reports: Employees by Department (pie), Employees by Location (bar) and Employee List (list).
- Employee Analytics Dashboard that can be shared with users, groups and roles.

Non-functional requirements: accuracy and data integrity, no duplicate records, administrator access to the required tables and ACLs.

3.3 Data Flow Diagram

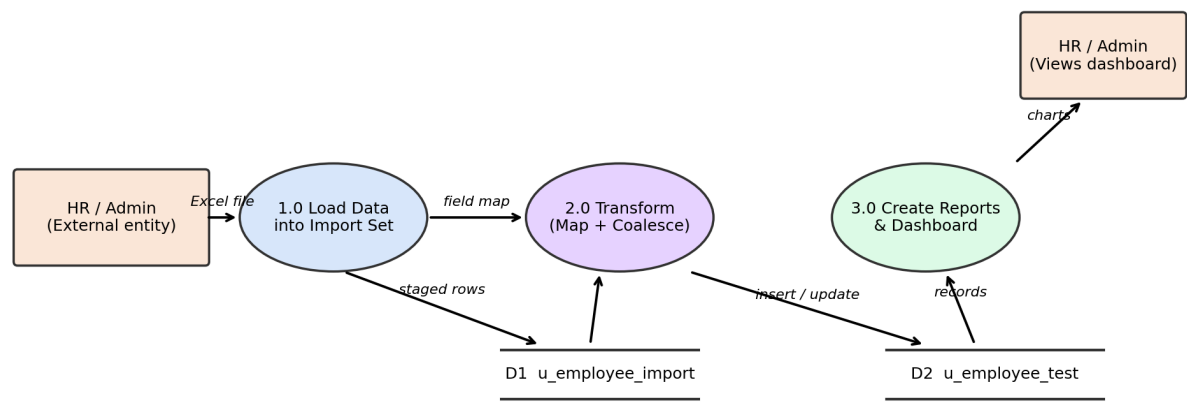


Figure 1: Data flow diagram of the employee data import

3.4 Technology Stack

Component	Technology
Platform	ServiceNow (Global application scope)
Source data	Google Sheets / Microsoft Excel (.xlsx)
Staging	Import Set table u_employee_import (Load Data)
Transformation	Transform Map with Coalesce
Target data	Custom table Employee Test (u_employee_test)
Reporting	ServiceNow Reports and Dashboards (Platform Analytics)

4. PROJECT DESIGN

4.1 Problem Solution Fit

Customer segment: HR teams, ServiceNow administrators and IT/data teams that receive bulk employee data in Excel. The problem root cause is that external spreadsheets are unstructured while ServiceNow needs structured table records; without field mapping and a unique key, imports are manual and repeated imports create duplicates. The solution (Import Set, Transform Map, Coalesce, reports and dashboard) fits the customer constraints of limited time, limited scripting skills and the need for accuracy.

4.2 Proposed Solution

Parameter	Description
Problem Statement	Manual, error-prone employee data entry and duplicate records on repeated imports.
Idea / Solution	Automated import through Import Set and Transform Map with Coalesce on Employee ID, plus reports and a dashboard.
Novelty / Uniqueness	Re-importing the same sheet gave 0 inserts and 4 ignored rows; a changed sheet gave 2 inserts and 2 updates. Entirely low-code.
Social Impact / Customer Satisfaction	Less manual work and fewer errors for HR/admin teams; reliable, up-to-date employee data.
Business Model	Implementation and data-migration service, support subscription, or a reusable import-and-dashboard package.
Scalability	Same pattern works for users, assets, incidents and CMDB items by changing the source file and field maps.

4.3 Solution Architecture

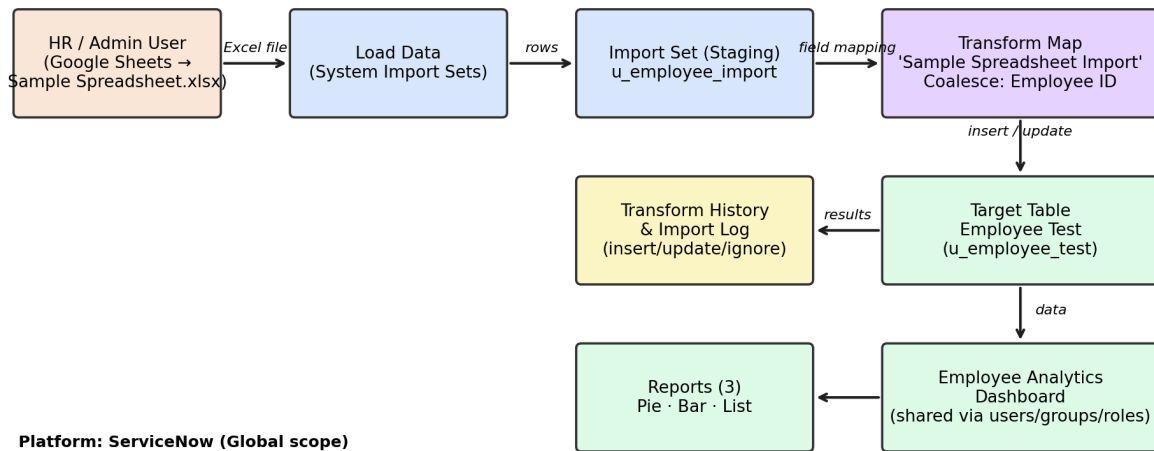


Figure 2: Architecture and data flow of the solution on ServiceNow

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Sprint	Epics	Story points	Duration	Dates
Sprint-1	Data Preparation, Table Setup, Data Import (USN-1 to USN-5)	12	6 days	19 Jan 2026 – 24 Jan 2026
Sprint-2	Duplicate Handling, Reports, Dashboard (USN -6 to USN-12)	16	6 days	26 Jan 2026 – 31 Jan 2026

Total story points = 12 + 16 = 28 over 2 sprints, so the velocity is 14 story points per sprint (average velocity $\approx$ 2.33 story points per day for a 6-day sprint).

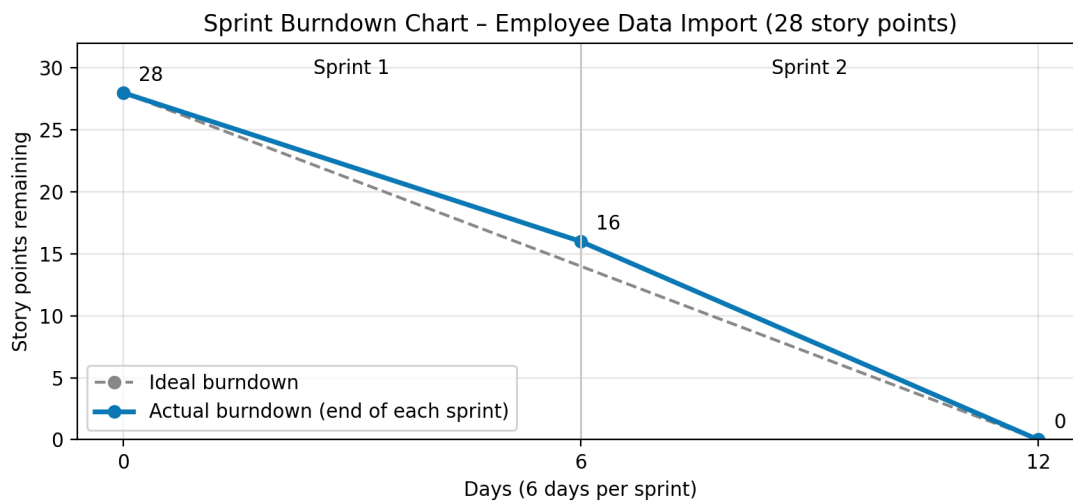


Figure 3: Sprint burndown chart

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Test	Result
First load of Sample Spreadsheet.xlsx	Processed 15, inserts 15, errors 0 (100% success)

Updated load (Coalesce on Employee ID)	Total 4: 2 inserts, 2 updates, 0 errors
Repeat load of the same sheet	Total 4: 0 inserts, 0 updates, 4 ignored – no duplicates
Reports and dashboard	Pie, bar and list reports run and display on the dashboard

User acceptance testing covered 11 test cases (table creation, import, transform, Coalesce, duplicate check, three reports and the dashboard); all passed. One low-severity issue was logged: the Department value appears as “Servicenow” instead of “ServiceNow” for two updated records.

7. RESULTS

7.1 Output Screenshots

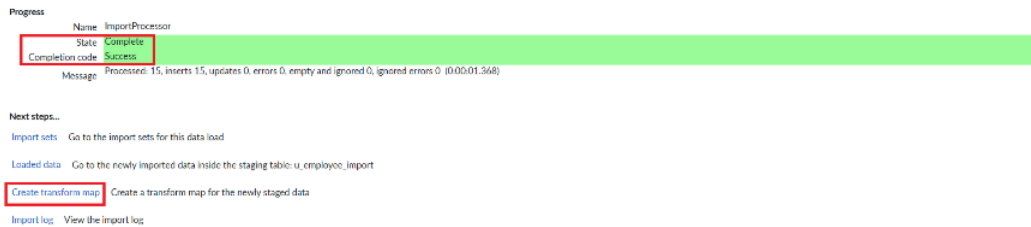


Figure 4: Import set load completed (processed 15, inserts 15)

Employee Tests Employee ID Search				
Employee ID	Employee Name	Email	Department	Location
SB-0003	Meghana Katoju	test3@gmail.com	ServiceNow	Hyderabad
SB-0001	Ratan Kumar	test1@gmail.com	ServiceNow	Hyderabad
SB-0015	Srinu Nandipalli	test15@gmail.com	ServiceNow	Hyderabad
SB-0011	Abhishek S	test11@gmail.com	Salesforce	Hyderabad
SB-0007	Yaswanth Nikku	test7@gmail.com	ServiceNow	Hyderabad
SB-0010	Ravi Teja	test10@gmail.com	ServiceNow	Hyderabad
SB-0006	Sushma Ganji	test6@gmail.com	ServiceNow	Hyderabad
SB-0002	Sandeep Guja	test2@gmail.com	ServiceNow	Hyderabad
SB-0014	Sushanth J	test14@gmail.com	Salesforce	Hyderabad
SB-0013	Mounika K	test13@gmail.com	Salesforce	Hyderabad
SB-0009	Praneeth S	test9@gmail.com	ServiceNow	Hyderabad
SB-0005	Bitla Sai Prasad	test5@gmail.com	ServiceNow	Hyderabad
SB-0008	Prashanth Reddy	test8@gmail.com	ServiceNow	Hyderabad
SB-0004	Ajay Kumar	test4@gmail.com	ServiceNow	Hyderabad
SB-0012	Rakesh Bhoomani	test12@gmail.com	Salesforce	Hyderabad

1 to 15 of 15

Figure 5: Employee Test table with 15 records

Transform Histories											
Started	State	Completed	Run time	Set	Import set table	Total	Inserts	Updates	Ignored	Skipped	Errors
2026-01-29 03:47:31	Complete	2026-01-29 03:47:38	1 Second	ISI 10010003	Employee Import [u_employee_import]	4	2	2	0	0	0

Figure 6: Transform history – 2 inserts and 2 updates

Transform Histories											
Started	State	Completed	Run time	Set	Import set table	Total	Inserts	Updates	Ignored	Skipped	Errors
2026-01-29 05:39:36	Complete	2026-01-29 05:39:36	0 Seconds	ISI 10010004	Employee Import [u_employee_import]	4	0	0	4	0	0

Figure 7: Transform history – repeat load ignored (0 inserts, 0 updates, 4 ignored)

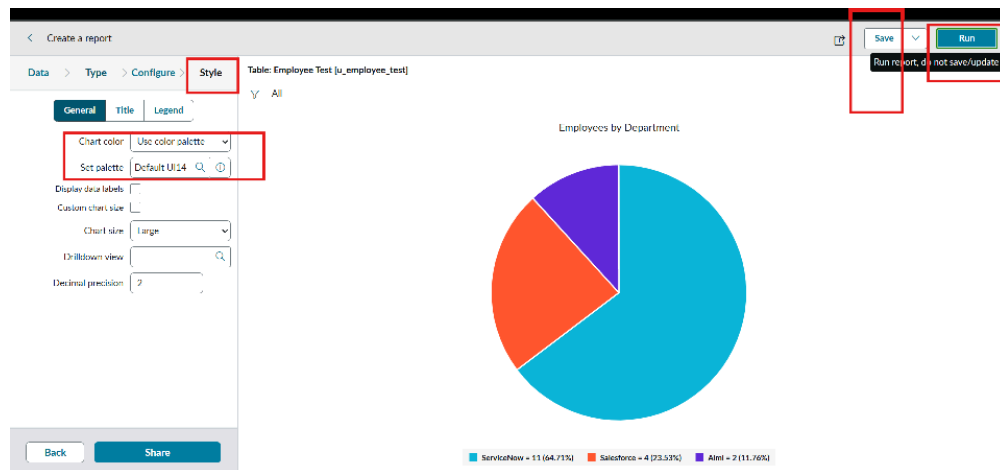


Figure 8: Employees by Department (pie chart)

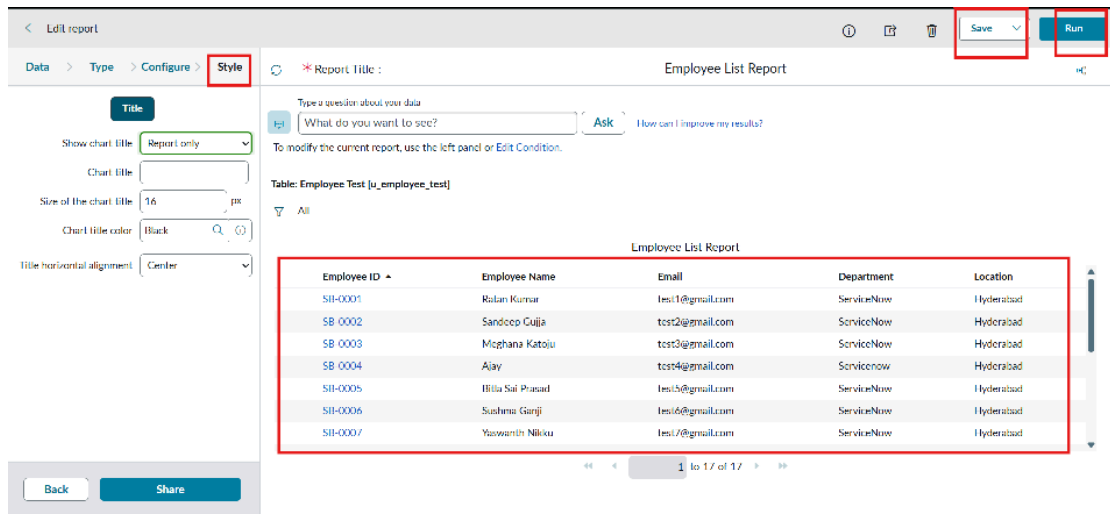


Figure 9: Employee List report



Figure 10: Employee Analytics Dashboard

8. ADVANTAGES & DISADVANTAGES

Advantages:

- Fast, accurate bulk import with no scripting.
- Coalesce prevents duplicates and updates existing records.
- Transform History gives clear counts of inserts, updates and ignored rows.
- Dashboard gives one place to monitor employee data.

- Reusable for other tables and data.

Disadvantages:

- The file must be uploaded manually; imports are not scheduled.
- Coalesce is set on one field only (Employee ID).
- Creating dashboards through the backend form required changing an ACL setting (Admin Overrides).
- Data values are copied as-is, so inconsistent text (for example "Servicenow") is not corrected.

9. CONCLUSION

The project implemented an employee data management solution using ServiceNow Import Sets and Transform Maps that keeps data accurate and consistent. Enabling Coalesce on Employee ID ensures repeated imports update existing records instead of creating duplicates. The reports and Employee Analytics Dashboard give administrators and HR teams a single view of employee data.

10. FUTURE SCOPE

- Schedule imports from a data source (JDBC, REST or SFTP) instead of manual file upload.
- Add a transform script to normalize text values such as department names.
- Add validation rules and notifications for import errors.
- Add more reports, such as new and updated records per import.
- Restrict dashboard access by role and extend the same pattern to other tables.

11. APPENDIX

Source Code: Not applicable (low-code configuration in ServiceNow).

Dataset: Sample Spreadsheet.xlsx – 15 employee records with Employee ID, Name, Email, Department and Location.

GitHub & Project Demo Link: